Introduction to Credit Risk and Applications in Python

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Course Overview

Duration: 9 sessions

Course Structure:

- *Python Foundations (2–3 sessions)* Data types, control flow, functions, and core libraries (NumPy, Pandas, Matplotlib)
- *Statistical Foundations (2–3 sessions)* Probability concepts, distributions, and basic data analysis
- *Credit Risk Modeling (3–5 sessions)* Default modeling, portfolio loss distributions, dependence structures, and risk measures

Prerequisites: Basic familiarity with Python and Git

Required Setup: Python, PyCharm, and Git installed before the course begins

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Learning Format

Lectures

- Introduction to Python programming and data analysis
- Development of credit risk models (default modeling, dependence, loss distributions)
- Hands-on implementation and simulation in Python
- Version control and collaboration using Git (basic workflow, repositories, and submission)

Assessment

Continuous Participation (25%)

- Attendance and active engagement during lectures
- Completion of in-class exercises and short assignments

Final Assignment (75%)

Timeline: 2 weeks

Task: Analyze a simulated credit portfolio dataset and develop a structured credit risk report

Deliverables:

- Written report (modeling approach, results, and interpretation)
- Python code (submitted via GitHub repository)
- 5–10 minute oral discussion of results